



RUN4

OPERATION MANUAL

Four-track Tape & Texture Machine

Edition 1 — 2026

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PART 01

Getting started

What RUN4 is, how it is laid out, and the habits that make the rest of the manual read easily.

How this manual is written

- Controls are named as they appear on screen. T1 to T4, Disc and Perform are the six destination buttons in one bar; Tape and Poly are the source control for the selected track, in the row beneath it.
- A tap is one short touch. A long-press is holding a control until its panel opens. Long-press is RUN4's route to assignment and detail throughout, and it never changes the value of the control you are holding.
- An arrow describes signal order, so Material -> Grain -> Tone -> Color -> Air -> mixer is the order audio actually travels in.
- Where material is a lookup rather than an explanation it is set as a table of trigger and action. The sentences around a table carry the detail that will not fit in a row.
- Every section here is also the in-app Help section of the same name. The app and this PDF are generated from one source, so they cannot disagree with each other.

What RUN4 is

- RUN4 is a four-track Tape & Texture Machine. Each track has a Material source, a device chain, mixer controls, sends, and performance state.
- The normal path is: choose a track, choose the Material mode, load or capture sound, shape it with devices, perform it with transport/macros/scenes, then export a master or stems.
- The four main areas are Tape, Poly, Disc, and Performance. Tape, Poly, and Disc are sound engines. Performance is the full-screen live stage for four waveforms, hands-on track controls, Sets, Rune Pad, and Punch-in FX.

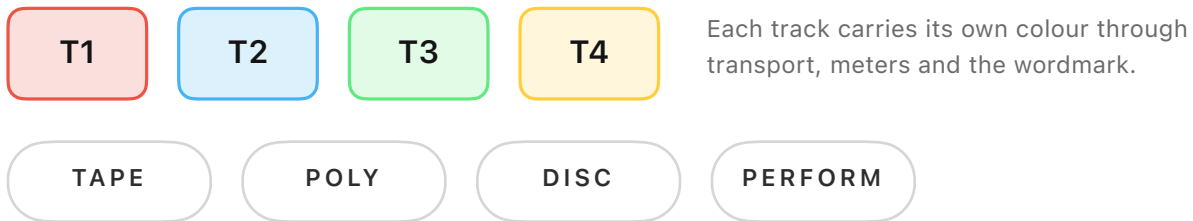
- The fixed device order is Material -> Grain -> Tone -> Color -> Air -> mixer/sends/master. Device defaults are transparent where possible, so moving a Mix or amount control is often what actually brings an effect into the sound.



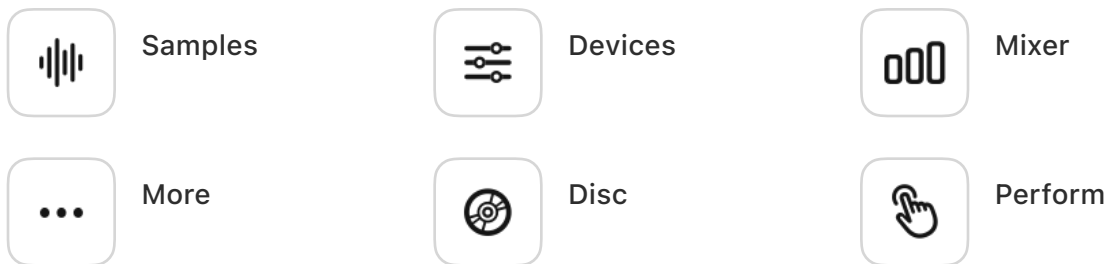
The fixed device order every track runs through

Navigation and tracks

- T1 to T4 select the active track. Track color follows the selected track and appears in transport, device activity, meters, and the reactive RUN4 bird/wordmark.
- One bar holds every destination: T1 to T4, then Disc for long streamed files and Perform for performance structure. It stays on screen in all six, so any destination is one tap from any other.
- Tape and Poly are not in that bar. They sit in the row beneath it and choose what the SELECTED track plays -- Tape for recording and looping, Poly for keyboard, sample, synth or AUv3 playback. Because Disc and Perform belong to the whole Set rather than to one track, that row is absent there.
- In the top row, the waveform button opens the Sample Browser, the sliders button opens Devices for the selected track, and the bar-chart button opens the Mixer. The ellipsis holds the current Project, Settings, and Save.
- On iPad the header also carries the project name, which you can rename in place. On iPhone that width goes to Set, Tempo and BPM instead, so the whole header stays on one row; open the ellipsis and choose Projects · [current name] to enter the same Project Manager.
- Inside Devices you reach Grain, Tone, Color, Air, and Mod for the selected track. Mod is a control source page rather than another stage in the audio chain. Settings, which contains this manual, is in the ellipsis menu rather than on a gear of its own.
- The active device strip lights when a device has moved away from its clean default. Clear inside a device resets that device for the selected track.

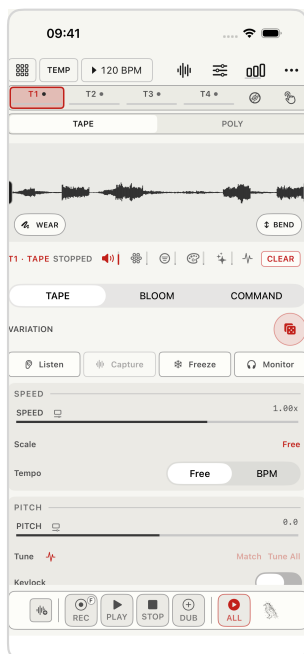


Track selectors and the mode row

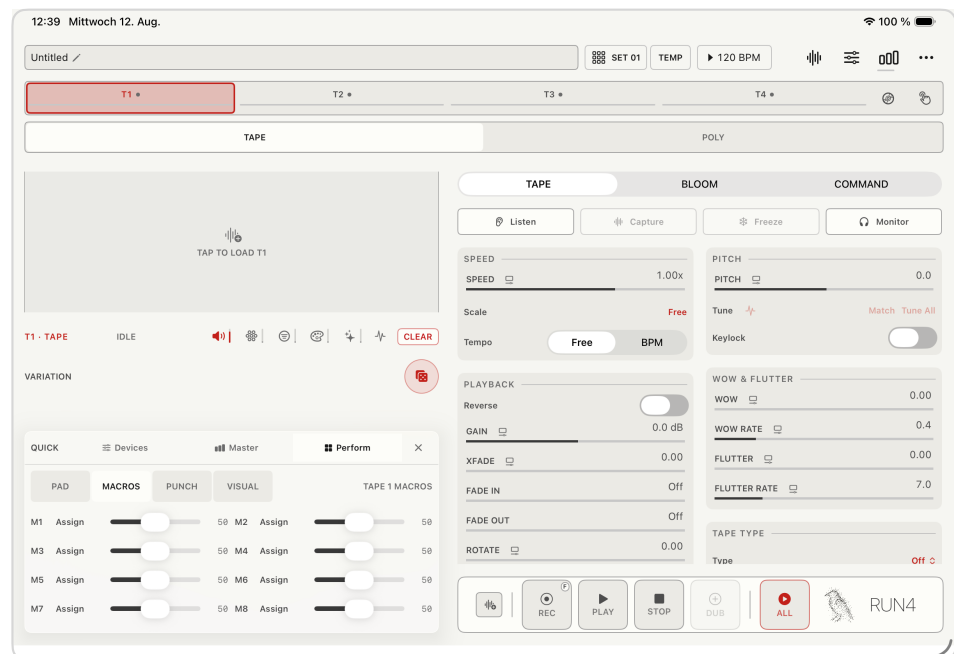


The destination bar, left to right: the four tracks, then Disc and Perform. They share one bar because that is the order a set is built in -- fill the tapes, set up the disc, then play. Tape and Poly sit below it and apply to whichever track is selected, so they are absent on Disc and Perform.

IPHONE



IPAD



Tape on both idioms. The header carries the project name, Set/Temp, the tempo and the tools; below it the destination bar (T1 to T4, Disc, Perform) and then the Tape/Poly source control for the selected track. iPad landscape lays the same controls out side by side instead of stacking them.

Projects and Auto Save

- Open the Project menu — the ellipsis in the top row, then Projects — to Save, Save As New, create an empty project, load a project, or save the current setup as a reusable template. Save is always explicit and available whether Auto Save is on or off.
- Projects > iCloud > Move to iCloud is an opt-in sync mode for the current project. It collects the used audio once; after that, Save and Auto Save keep the same cloud project current across iPhone and iPad. Open Cloud downloads its latest state into RUN4's shared local storage, where the standalone app and AUv3 can both use it. The iCloud and Transfer sections start collapsed to keep Projects compact on iPhone.
- Make Local stops future iCloud updates for that project and keeps both the local project and its last cloud copy. If two devices edit the same revision, RUN4 stops before overwriting either version and asks you to open the latest cloud project.
- Transfer > Export Project collects the project and all audio it uses into one .run4project package. Save it to Files or iCloud Drive, then choose Import Project on another iPhone or iPad to continue working there.
- Projects preserve the identity and saved document state of hosted AUv3 instruments and effects, but they do not include the plug-ins themselves. Install the same AUv3 on the other device, then open the project in the standalone RUN4 app to restore it; RUN4's own AUv3 cannot host another AUv3. Restoration of plug-in-managed presets, samples, and other external content depends on that plug-in and its installed version.
- Auto Save is on by default in Settings > General. It keeps the open project updated while you work, including when you leave RUN4.
- When Auto Save is off, RUN4 writes project files only when you choose Save or Save As New. Opening another project does not save the current one; starting a new project or using a template warns before unsaved changes are discarded.

Undo and Redo

- Destructive actions are checkpointed: pressing REC on a tape that already holds a take, Clear, the Reset choices, and replacing a loaded sample. Open the Undo menu in the top bar to step back. A restored checkpoint returns the sample, the loop in and out points, Pitch, and the rest of that tape's settings exactly as they were.
- Undo restores a whole project state rather than a single control, so stepping back also reverts changes made after that action. Redo brings them back: the state before each Undo is captured first, so moving backward and forward never loses anything.

- The history holds twelve destructive actions and survives normal editing and Auto Save. Opening another project or starting a new one clears it.
- DUB is deliberately not covered. An overdub layers into the existing take and rewrites its file in place, so material from before the overdub cannot be restored. Record onto another track when you want to keep the original.
- Parameter edits have their own history. Undo Edit and Redo Edit, in the same menu, step through recent control changes: the Grain, Tone, Color, and Air devices, Tape's playback parameters, and Poly. One drag is one step no matter how many values it passed through on the way, and a drag that ends where it started records nothing.
- That history holds forty-eight edits, belongs to the scene it was made in, and lasts for the session. It puts the controls back in place: nothing is reloaded and a playing tape keeps playing, so an Undo during a set changes only the parameter it undoes.
- Smaller, local undos stay separate: the die in Tape, Bloom, and Command has its own one-step Undo per section, and the Tune row has its own revert buttons.

Long-press shortcuts

- Rec Grid quantizes Record, Overdub and Play All to RUN4's own tempo grid when no external clock is running; the Metronome clicks that same grid.
- When a parameter is assigned to one of the eight macros M1 to M8, its small M label appears beside the parameter name; tap that label to open the matching modulator directly.

LONG-PRESS TARGETS

BPM display, anywhere in RUN4	Timing & Sync: MIDI Clock In and Out, Ableton Link, Sync Quantize, Rec Grid, Metronome
Any routable parameter or selector	Assign or remove a modulation source, or map it to a track Macro or Rune Pad
Poly pattern step	That step's detailed editor: source, Kit slot and playback, Length, Velocity, Gate, Pitch, Probability, Repeat, Micro, and Mute
Quick device icon	That device's full panel

PART 02

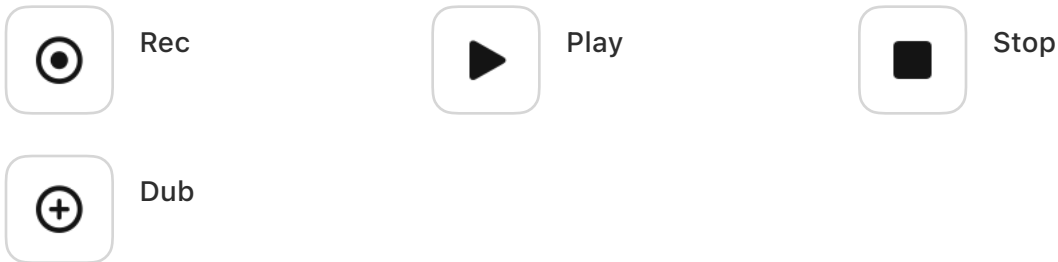
Tape

The four tape tracks: recording, loops, touch tools, tuning, Bloom, and the Command Sequencer.

Tape mode overview

- Tape records audio into a loop, plays it back, overdubs into it, and can resample the internal mix through Listen/Capture. Reach it from a T1–T4 chip, then Tape in the source row beneath it. Tape is the quickest mode for live sampling and transforming captured audio.
- The die in Tape, Bloom, and Command rolls a variation for the visible section. After a roll, a visible Undo button restores it. Undo is one step per section; a new roll or a manual change to that section starts a new state.
- REC records into the selected tape. PLAY starts playback using the Tape Start setting: Restart, Resume, or Align. STOP stops the selected tape. DUB overdubs onto existing tape material. ALL starts or stops all track transports together. Long-press a top T1–T4 chip to toggle Solo in Place directly: other tracks fade out but keep running in sync, and return at their current positions when the final Solo is removed. Several tracks can be soloed together.
- Clear offers destructive reset choices. Only Tape clears the tape material/state. Tape + Devices also resets Grain, Tone, Color, and Air on that track.
- Listen arms the internal always-listening buffer. Capture copies the recent internal mix into the tape without needing to route speaker output into the microphone.
- Freeze holds the tape player at the current moment for sustained texture and performance gestures.
- Two fixed touch chips sit on a loaded waveform. The Tape touch tools section below explains persistent position painting, momentary gestures, dual-gesture performance, and the Touch Presets that configure the two slots.

- Tapping an empty waveform opens Samples with that tape already set as the load destination. A tape that holds a take keeps the tap for panning, zooming, and dragging the loop handles; the load target appears only while the window is genuinely empty or holds a take too quiet to be heard.
- The waveform draws quiet material larger than it truly is, so a faint take stays readable instead of collapsing into a flat line. Louder material is drawn at its real amplitude, and Gain still visibly scales the picture.

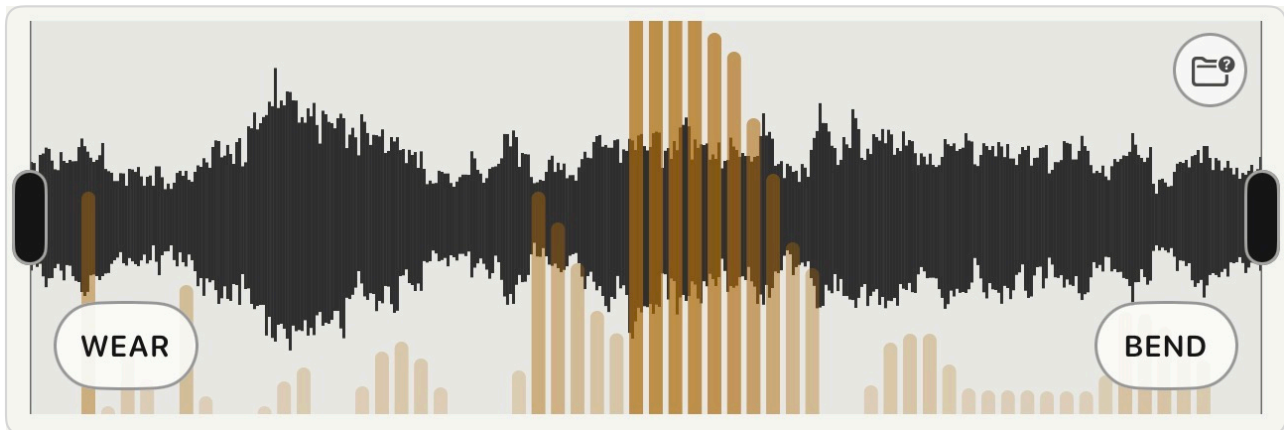


Transport controls

Tape touch tools

- Two fixed chips sit on every loaded Tape waveform — reach them from a T1–T4 chip, then Tape. Normally the left chip paints a persistent behavior onto positions in the recording and the right chip performs a momentary gesture. The selected tools are stored with that Tape, so the two visible slots never need to grow.
- When the left chip is a Paint tool, tap it to enter Paint mode, then draw across the waveform. Horizontal position chooses the physical point on the Tape. Drawing higher applies more; drawing near the bottom cleans that position. Tap the chip again to leave Paint mode. Long-press it to choose Wear, Melt, or Bloom Paint, clear the current map, switch to two gestures, or load a Touch Preset.
- Wear adds immediate position-locked drag, instability, and sparse oxide holes. It works even when Tape Character Wear Rate is at zero; when cumulative Wear is active, painted marks also become the areas that age fastest. Melt darkens, ages, and slows the marked material as it passes the head. Bloom Paint brings in the existing six-head Bloom only at the painted positions without changing the global Bloom switch or its authored head setup.

- Drag a Gesture chip to perform its selected movement. The label shows the live amount while the finger is down. Long-press without moving to choose another gesture; the chooser remains available while the Tape is playing.
- Bend moves Tape speed by up to three semitones. Releasing coasts smoothly back to the authored Speed; a larger bend takes slightly longer to return, while Elasticity adds lag, overshoot, and settling. Brake combines a downward motor pull with temporary Melt. Catch repeats the current fragment from $\times 2$ to $\times 8$.
- Fold uses vertical travel so it works equally well at the waveform edge: drag upward for reverse Fold or downward for forward Fold. Distance selects the musical divisions $\times 2$, $\times 4$, $\times 8$, and $\times 16$. Squeeze also works in either vertical direction and narrows the playback window around the point caught when the gesture begins. Melt temporarily increases Tape Character Melt.
- Bloom Hold spreads the existing six-head Bloom only while the finger stays down; horizontal position changes its motion. Bloom Throw gives the heads momentum so they coast and fade after release. Rub is a linear vertical 0–100% control: distance sets the amount directly, adding drag, wear, snags, and brief holes, and the value remains stable until release.
- Choose Use Two Gestures from the left chip to turn it into a second momentary gesture slot. Long-press either chip to choose its gesture independently, then use two fingers to perform both at once—for example Bend on the left and Bloom Throw on the right. Long-press either gesture and choose Restore Paint on Left to return immediately to Paint + Gesture. Choices that share the same playback state are disabled so releasing one finger cannot cancel the other gesture.
- Paint maps remain attached to the Tape while two gestures are visible. Performance gestures are temporary and do not overwrite the visible Tape controls. Touch Presets are only shortcuts, not extra modes: Tape Hands selects Wear Paint + Bend, Bloom Hands selects Bloom Paint + Bloom Hold, and Broken Tape selects Wear Paint + Catch.



Tape Touch in use: Wear is painted into positions on the waveform while Bend remains ready as the momentary gesture.

Tape recording and fixed loops

- A normal tap on Tape REC — reach Tape from a T1–T4 chip, then Tape — uses the currently selected capture setup. Tap the waveform-plus control inside the transport box to open its persistent capture panel.
- Record captures and stops at the end. Record & Play captures and immediately starts the new Tape as a loop.
- Length can be Free or a fixed 1, 2, 4, or 8 bars. Fixed length follows the current RUN4 tempo and 4/4 bar grid; for example, 2 bars at 120 BPM record for 4 seconds. A fixed capture starts directly at its record edge and therefore bypasses the input Level Trigger; Free capture keeps the Level Trigger behavior. During Free capture the waveform HUD shows elapsed time; during a fixed Tape or Poly/AUv3 capture it shows the chosen bar count and remaining time.
- Count-in can be Off or 1 Bar. With 1 Bar, READY counts 4 · 3 · 2 · 1 on the current grid before the recorder starts. The four clicks run independently of the saved Metronome On/Off and activity mode, using its selected click level; Stop or a second REC cancels the preparation immediately.
- The capture panel also includes the Metronome and a continuous click-level fader. It remains open while you adjust the level. The click follows quarter notes and accents beat 1. In the standalone app it runs outside Tape and Export. Inside the AUv3 it must leave on RUN4's output bus to be audible, so a host recording that plug-in can include it; RUN4's own internal Tape capture point remains clean.

- Rec Grid and fixed capture length are related but separate: Rec Grid chooses when a transport action begins, while Length chooses when a fixed recording ends.

Tape parameters

- Reach these from a T1–T4 chip, then Tape, under the Tape tab rather than Bloom or Command.
- Listen/Capture keeps an 8-second internal mix window for quick resampling.
- Speed is varispeed playback from 0.0625x to 4x. It changes timing and pitch unless Keylock is enabled.
- Tempo switches Speed between Free behavior and BPM-aware behavior. In BPM mode, Tape BPM tells RUN4 what tempo the captured tape originally represents.
- Pitch transposes tape playback in semitones from -24 to +24.
- Scale chooses pitch snapping for tape speed/pitch behavior: Free, Chromatic, or scale-based choices. When Scale is active, Root chooses the tonal center.
- Start Grid and Length Grid appear in BPM mode. They quantize loop start and loop length to musical values from 8 bars down to 1/16, or Free.
- Keylock keeps pitch more stable while Speed changes. Reverse flips playback direction.
- Gain sets tape output level. Free waveform trims settle onto a nearby zero crossing, while a 3 ms safety blend protects hard source and tempo-grid edges without changing recording length. XFade extends that loop-wrap blend up to 500 ms, useful for pads and clean sustained loops. Rotate offsets the loop/read position. Glide smooths motion between playback changes.
- Shadow blends in one hidden companion playhead rather than showing another head on the waveform. It follows the same loop while a stored random seed drives slow, bounded position, rate, and pan deviations that approach a target, overshoot, and settle instead of wobbling like unrelated LFOs. The same project therefore recalls the same Shadow character. Level compensation keeps the control about depth rather than loudness; Bloom and Freeze take priority, and Shadow fades back quietly when normal Tape playback returns.
- Fade In and Fade Out shape each loop pass and are Off by default: loaded or recorded material is never reshaped until you ask for it. Full range is half the loop, so both at maximum form a single triangular swell. Setting only one of the two still leaves the loop wrap click-free, because the opposite edge is given a short automatic minimum.

- The waveform draws Fade In and Fade Out at the length you set, and draws nothing when they are Off. Click-free looping is playback behaviour rather than an edit to your material, so it is not drawn.
- SOS controls sound-on-sound retention during overdub: high values keep previous material longer, low values replace it faster.
- Wow and Wow Rate add slower tape-speed drift. Flutter and Flutter Rate add faster instability.
- Tape Type is Off by default, so existing tapes keep their original sound. Reel, Sixties, 4-Track, Cassette, Video, and Dictaphone each add a calibrated medium response through bandwidth, dynamics, a subtle noise floor where appropriate, and their own quiet mechanical movement. Reel has only a very small soft-saturation bed for body; Sixties, 4-Track, and Dictaphone are the deliberately coloured types, while Cassette and Video stay clean.
- Tape Type never rewrites the front-panel Wow, Flutter, Failure, Noise, or Splice values. Those remain your additive performance controls: choose a medium first, then add as much motion, wear, hiss, or damage as the moment needs.
- Tape Character Age adds instant colour and bandwidth loss. Drift adds slow musical instability, Elasticity changes how the motor settles after speed movement, Dust creates short oxide holes, and Melt darkens, loosens, and slows the Tape.
- Tape Character Wear Rate controls how quickly the active loop ages and starts at zero. A looping Tape ages a little after each completed pass only while Wear Rate is raised; Stop or returning Wear Rate to zero freezes the reached state instead of resetting it. Accumulated shows the wear reached inside the active loop, stores that history with the Tape, and Restore removes it at any time. Wear Paint is optional and only makes its marked positions age faster. RUN4 never rewrites the source audio—export a Stem when you want to print the worn sound.
- Degrade Failure adds wear, pitch snags, crinkles, and dropouts. Degrade Noise adds hiss. Noise Color changes the character of that hiss.
- The die beside Tape Variation creates a restrained new tape character: Speed, Pitch, playback movement, Wow, and Flutter change, while your source, gain, loop, routing, and project tempo stay untouched. Long-press that die to restore the exact pre-roll settings.

Tape Tune, Match, and Tune All

- The Tune row is for bringing recorded Tapes into a musical relationship; reach it from a T1–T4 chip, then Tape, in the Tuning card beneath Pitch. It is available when the selected track has a tape take; key analysis works from the current loop region and reports the audible key, including the current Tape pitch and non-Keylocked speed shift.
- Tap the waveform/analyze button to detect the selected Tape's key. The result is a guide, not a forced correction: sparse, noisy, or highly atonal recordings can return No key.
- Match analyzes the selected Tape and the other recorded Tapes, then offers pitch-match choices. Choosing one changes the selected Tape's Pitch and switches its pitch snapping to Chromatic; the audio file itself is not rewritten. Use the curved-arrow button to revert the latest Match on that Tape.
- Tapping Match again reopens the same list instead of analyzing a second time, so Best, Fifth, Darker, and Brighter can be tried against each other; the choice in effect is ticked, and picking another replaces it rather than stacking a second shift. Use the analyze button when you want a genuinely fresh analysis, for example after retrimming the loop.
- Tune All > To T1–T4 uses the chosen recorded Tape as the reference and applies the best pitch match to every other recorded Tape it can analyze. The circular curved-arrow button reverts the last Tune All gesture as one action, provided those affected tracks have not been changed since.

Tape Bloom

- Bloom turns one tape read head into a small cloud of six read heads. Reach it from a T1–T4 chip, then Tape, then the Bloom tab. It is useful for chorused loops, smeared harmonies, clouds, and animated tape swarms.
- Spread separates the read heads around the tape. Motion animates the read-head positions. Stereo widens the heads across the field.
- Warble and Warble Rate add pitch/time instability to the Bloom cloud. Degrade adds tape wear specifically to Bloom behavior.
- Each Head has Direction, Level, Pitch, Speed, and Pan. Direction can be Stop, Fwd, or Rev. Pitch is in semitones. Speed changes that head's playback rate.
- Reset Global restores Bloom's shared controls. Reset Heads restores all six head-level settings.

- The die beside Bloom · All Heads turns Bloom on and creates one balanced six-head cloud, rather than randomizing each head into unrelated extremes. Long-press that die to restore the previous Bloom state, including every head.

Command Sequencer

- Command is a 16-step performance sequencer for tape actions. Reach it from a T1–T4 chip, then Tape, then the Command tab. It can jump, reverse, retrigger, squeeze loops, stutter, gate, hold, nudge, spread Bloom, or trigger Memories.
- BPM is the shared master tempo. Editing it here changes the same project tempo used by Link, MIDI clock, and synced behavior.
- Pattern selects A, B, C, or D. Chain lets four pattern slots define a performance order. Seed Lock keeps random/probability behavior repeatable; New Seed creates a new variation.
- Each step has a command, an optional value, and Probability. Probability at 1 always fires; lower values make the step conditional.
- Jump uses Position. Rate uses Rate. Spread changes Bloom spread. Squeeze changes loop size. Stutter uses Repeats. Nudge uses Offset.
- Memory steps choose a stored Memory and can pitch it chromatically from -24 to +24 semitones. Level, Fade In, and Fade Out shape each triggered snippet. A Memory used by a Command step stays protected from automatic Memory-bank eviction. Clear Pattern resets the active pattern.
- The die beside Command · Pattern fills the selected pattern with sparse, playable tape gestures. It never creates Memory steps, so it cannot unexpectedly trigger stored audio. Long-press that die to restore the entire previous selected pattern.

PART 03

Poly and Disc

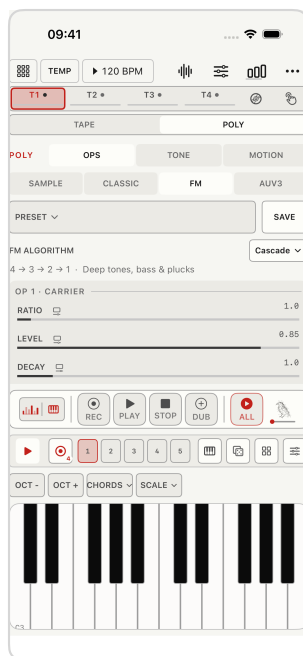
Keyboard, sample and synth playback, and long streamed files.

Poly mode overview

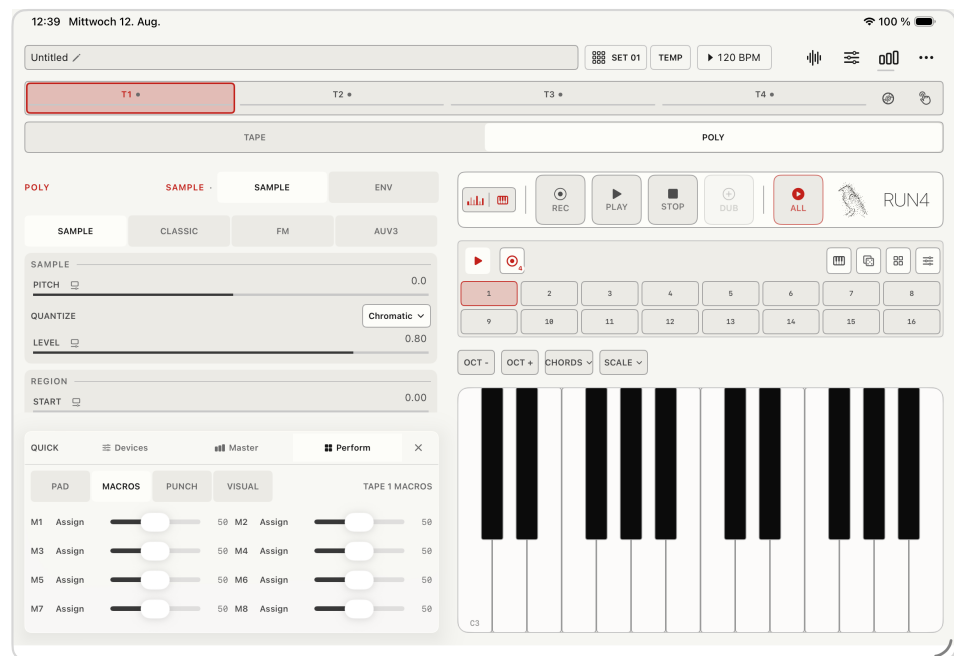
- Poly plays the selected track as a polyphonic instrument. Reach it from a T1–T4 chip, then Poly in the source row beneath it. It can use a loaded sample, the internal synth, or a hosted AUv3 instrument in the standalone app.
- The keyboard and MIDI share the same voice path. Octave buttons shift the playable range; the current base octave is shown at the bottom of its C key on the keyboard.
- Choose Sample, Classic, FM, or AUv3 directly at the top of Poly. Classic and FM are RUN4's internal synth engines; in the AUv3 extension build, AUv3 hosting is hidden because iOS does not allow an AUv3 extension to host other AUv3 instruments.
- The keyboard and step sequencer can be shown or hidden independently with the two display buttons inside the Poly transport box. Hiding them also collapses their keyboard/scale controls and returns the freed space to the selected Sample, Synth, or AUv3 editor.
- OCT –, OCT +, CHORDS, and SCALE share one control row. Tap CHORDS or SCALE to open its menu.
- From T1 to T4 selects a Tape as the musical reference. Match analyzes that Tape and applies its detected root and major/minor key to the active keyboard mode: SCALE receives a matching guide, while CHORDS receives the corresponding diatonic chord map. It does not change the source Tape. The CHORDS and SCALE menus can lock performance input to the active chord map or matched scale. A lock keeps the piano layout intact, dims unavailable keys, and filters live MIDI/QWERTY notes as well as on-screen playing.
- Poly's step strip -- sixteen steps to a bank, up to sixty-four in a pattern -- is a compact note or Kit input for the selected Poly source and for printing ideas into Tape. It does not arrange Tape tracks or build songs and is intentionally not a complete song or multitrack sequencer. The Poly Step Sequencer section below covers its steps, long events, Kit behavior, and pattern timing.

- Long-press the main Poly REC button to select Free or a fixed bar length, Count-in Off or 1 Bar, and the Metronome settings. Tap REC to capture the live Poly or hosted AUv3 output into the paired Tape. The small record-circle in the sequencer remains a simple Print-to-Tape action and uses those same settings. Poly shows READY during the preparation bar, then a normal time countdown for fixed captures and elapsed time for Free. The dice randomizes performance values. Kit Presets save and recall the eight slot choices only, so the same Kit can carry different patterns, or one pattern can be re-voiced by loading another Kit under it.
- Sample, Synth, and hosted AUv3 instruments can be performed or printed directly into the selected Tape. Fixed-bar printing is useful for tempo-aware AUv3 grooveboxes because RUN4 supplies the project transport and closes the Tape on the selected bar boundary.

IPHONE



IPAD



Poly on both idioms. Keyboard, pattern, and sample or synth controls for the selected track; on iPad the keyboard and the pattern grid sit beside the parameters rather than below them.

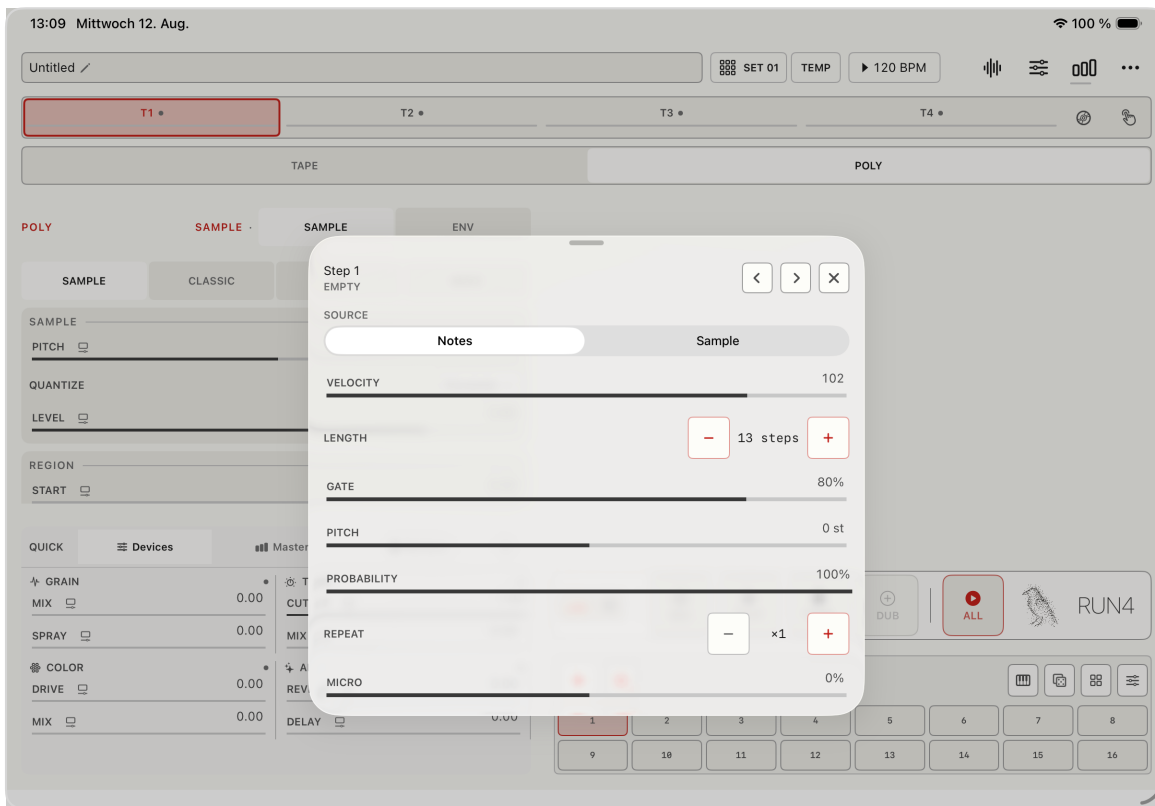
Poly Step Sequencer

- Poly's step strip is a compact note and Kit sequencer for the selected track, reached from a T1–T4 chip, then Poly. Sixteen steps are shown at a time and a pattern can contain up to sixty-four. It plays the current Poly sample, synth, hosted AUv3 instrument, or one of the eight Kit slots; it does not arrange the four Tape tracks.

- Tap or long-press any step to select it and open the inline Step editor. Its Source, Event, Pitch, and Behavior cards keep Kit slot and playback, Length, Velocity, Gate, Pitch, Probability, Repeat, Micro timing, and Mute clearly separated. A step plays either notes or one Kit slot, never both; muting now stays explicit in the editor instead of being hidden behind a quick step tap.
- Every step shows its Velocity as a fill inside the cell, so the dynamics of a whole pattern are readable without opening a step. The fill is in the track colour where the step actually sounds and neutral grey where it is empty or muted.
- On iPad's expanded grid, drag up or down on a step to set its Velocity directly, and keep the finger down while moving sideways to draw a ramp across the pattern in one stroke; empty steps accept a value too, so a curve can be shaped before the notes are there, and the drag never creates a note, only the number moves. On iPhone's compact strip, that horizontal motion scrolls to reach steps beyond the screen instead, so Velocity there is set in the Step editor.
- Probability appears as a fine line along the bottom of a cell, and only when it is below one hundred percent. At full probability every step would carry the same line, which says nothing, so the line appearing is the signal.
- Length lets an event continue across later grid steps. It does not occupy or lock those steps: every later event remains independent and can trigger while the earlier note or sample is still sounding. A line along the bottom of the grid shows the complete span.
- Gate controls how much of that Length is heard for notes and Gate-mode Kit slots. A three-step event at 80% therefore sounds for 2.4 steps. Repeat divides the complete Length into ratcheted events rather than repeating only inside the first grid step.
- Resolution defines the musical duration of one step. It ranges from 4 Bars, 2 Bars, and 1 Bar through 1/2, 1/4, 1/8, 1/16, 1/32, and 1/64. Pattern length and Resolution multiply: sixteen steps at 1 Bar create a sixteen-bar cycle.
- Each K1–K8 slot has a playback mode. Gate uses the step's Length and Gate; One-shot plays the complete source sample even when its step ends. Other Kit steps can trigger and overlap while it plays. Existing projects open in Gate mode, preserving their authored timing.
- Pattern Stop is also the Kit Kill All: it cancels pending hits and immediately silences every Kit sample started by the pattern, including One-shots. RUN4 Stop All additionally silences the complete performance.

- The Pattern workspace opens in Poly's upper editor instead of covering the workspace with a sheet. Its Step, Tools, and Kit tabs keep the sequencer and keyboard visible and playable below. Step edits the selected event; Tools groups Pattern setup, Sound to Pattern, Transform, Euclidean creation, and targeted edits into the same parameter-card language as Poly and Tape; Kit loads K1–K8 and manages Kit presets without a modal panel.
- Sound to Pattern is the starting point: choose Notes, All, or K1–K8. Notes can be entered immediately with the still-visible keyboard and Step Entry. A Kit target exposes all eight slots plus Load Sample, so an empty project can load a sound and create its first Euclidean rhythm without visiting another setup screen first.
- Reverse mirrors the active steps end to end. $\times 2$ doubles the pattern length and copies the current cycle into the new second half, which is the quickest route from an eight-step idea to a sixteen-step one with a varied tail. $\div 2$ halves the pattern and keeps the first half; the steps beyond the new end are retained exactly as a manual Length change leaves them.
- Swing offsets every off-beat step, from Straight through 54%, 58%, and 62% to 66%, which is the familiar triplet feel. It is written into the steps' own Micro timing rather than stored as a separate pattern setting, so a swung pattern stays hand-editable and a note added later arrives already in the pocket. Because the control reads its value back out of the steps, moving a single hit by hand is visible there rather than hidden behind a setting the pattern no longer follows.
- Humanize drifts Velocity by up to 12% and Micro timing by up to 6% around the values already authored, unlike the dice, which draws fresh values across the full musical range. Gate and Probability are deliberately left untouched, and rests are never varied. A short Applied confirmation makes the otherwise subtle edit visible.
- The Target chooses what the second group acts on: All, Notes, or a single Kit slot from K1 to K8. This is how one sound can be moved, filled, or emptied against everything else in the pattern. A step holds one event -- notes or one Kit slot, never both -- so a Kit Target is the closest thing this sequencer has to a lane.
- Rotate Left and Rotate Right move complete steps with their assignments and locks; a long event landing at the pattern boundary is shortened to fit. With a Notes or Kit Target only that source moves. Because a step holds one event, an event whose destination is already taken by a different source keeps its own step instead of overwriting it. Nothing is ever lost this way, and a rotation with every destination blocked simply makes no change.

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- Euclidean distributes one source across Hits and Rotate. With a Kit Target it also works on an empty pattern, so five hits of K4 can be placed without a populated step to copy from, and it reuses whatever locks the pattern already gives that slot. A slot with no sample loaded and no existing use is refused rather than scattering silent triggers. Apply uses the displayed values; Randomize chooses a contained new hit count and rotation, applies it immediately, and updates both steppers. Events from other sources are never overwritten; matching off-pulse events are only muted, so their details remain recoverable.
 - Clear empties every step the Target matches, from the complete pattern down to a single Kit slot.
 - Undo restores the pattern as it was immediately before the last Pattern Tool, whether that was Rotate, Reverse, $\times 2$, $\div 2$, Swing, Humanize, Euclidean, Clear, or a performance randomization. It is one step rather than a history, and any hand edit afterwards releases it so it can never discard newer work. The same action also sits in the pattern dice menu.
 - There is currently no Loop or Retrigger/Continue mode per Kit slot. Use Gate for authored duration or One-shot for the complete file.
 - Step Entry writes played MIDI, onscreen, or QWERTY notes into the selected step. Auto Advance moves to the following step after entry.



A long event. Its duration is visible, while later steps remain independently playable.

Poly sample controls

- A live field sits above the controls and draws the loaded sample — reach it from a T1–T4 chip, then Poly, then Sample at the top of Poly. The playable region set by Start and Length is shaded over the waveform, so those two values point at something instead of standing alone as numbers. While a note sounds, a thin line runs through the field at the read position of the newest voice.
- With no sample loaded the field stays empty and reads Load Sample. Tapping it anywhere opens the Sample Browser, which is the shortest way to fill the track.
- Pitch sets the sample pitch offset in semitones. Pitch Quantize chooses Chromatic semitone snapping or Free continuous detune.
- Start chooses the sample start point. Length chooses how much of the sample is playable from Start.
- Level sets per-voice output level. Loop chooses the loop start point inside the playable region. XFade smooths the sample loop transition for up to 500 ms.

- Loop Mode Fwd loops forward from end back to loop start. Flip ping-pongs forward and backward.
- Attack, Decay, Sustain, and Curve shape the amp envelope. Decay currently also acts as release for the sample voice.
- Velocity controls how much MIDI or keyboard velocity affects loudness.
- Fltr Amount applies a filter envelope to the voice. Fltr Shape changes how that filter envelope follows or differs from the amp envelope.

Poly synth and AUv3

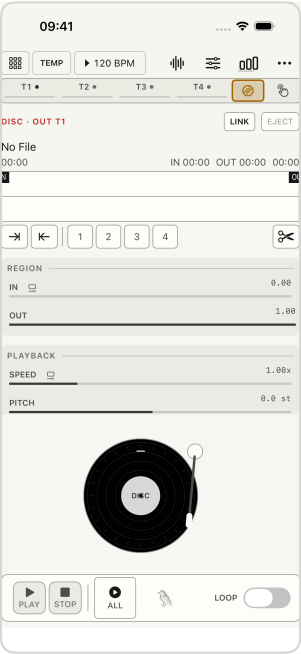
- Synth mode offers the Classic subtractive engine and the four-operator FM engine. Reach either from a T1–T4 chip, then Poly, then Classic or FM at the top of Poly; AUv3 sits beside them for hosted instruments. Presets are scoped to the selected engine and can be selected or saved.
- The engine selector sits in the fixed header beside POLY, so what is sounding stays visible while the controls scroll. Its named sections sit directly above the cards they switch: Sound, Filter, and Motion for Classic, Ops, Tone, and Motion for FM, and Source and Env for the sample player.
- Classic and FM each have a live field above their controls. It draws one note event: the amp envelope as the outline, filled with what the engine actually makes. Classic shows the selected waveform; FM renders its operator pair, so moving Ratio or an operator Level visibly changes the curve. Attack, Decay, and Release share the width in proportion to their times, which makes a long attack visibly eat the rest of the event. The amp envelope is shared by both engines, so its outline is the same in each.
- In Classic, Waveform chooses Sine, Triangle, Saw, or Square. In FM, Cascade, Merge, Dual, and Split select the operator routing.
- Synth Filter has Cutoff and Resonance. Cutoff is mapped musically from low to high frequency; Resonance adds emphasis without intentional self-oscillation.
- Amp Env has Attack, Decay, Sustain, and Release. Filter Env has Env Amount plus Flt Attack, Flt Decay, Flt Sustain, and Flt Release.
- LFO has Shape, Rate, Pitch Amount, and Cutoff Amt. Pitch Amount reaches up to about an octave of peak pitch movement; Cutoff Amt moves the filter over several octaves.
- AUv3 mode scans available Music Device instruments. Select loads an instrument, Edit Plugin opens its editor when available, and Remove AUv3 unloads it.

- Hosted AUv3 instruments receive RUN4's tempo, beat position, time signature, and play/stop state. This lets transport-aware grooveboxes and sequencers start, stop, and print against the same musical grid.
- When supported by the plugin, its editor can open Docked inside Poly, Compact, or Full Screen. RUN4 keeps the selected Poly page and editor presentation when the device rotates.

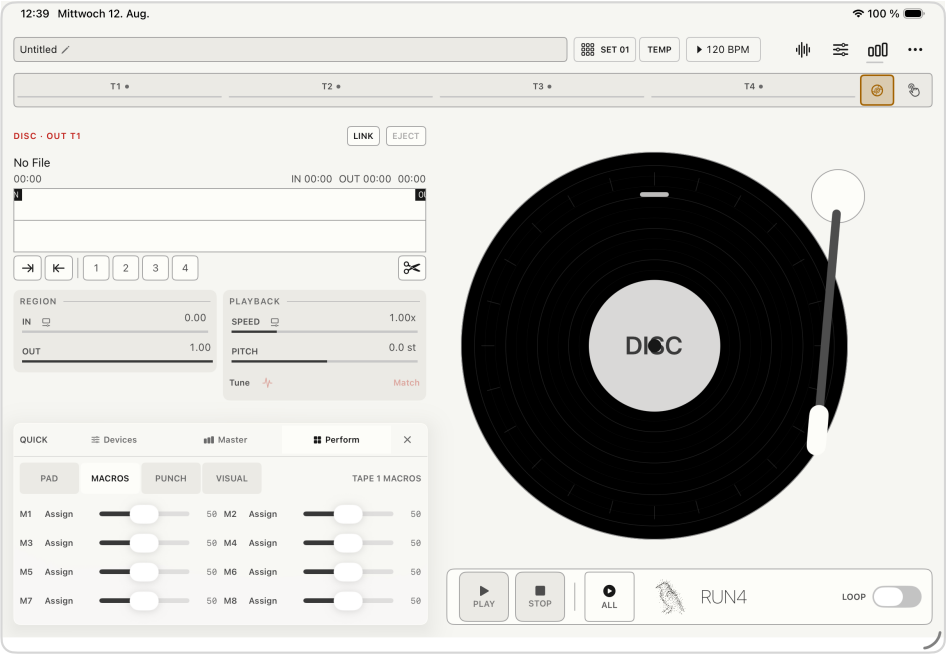
Disc mode

- Disc is one shared performance deck for the Set, reached from the Disc chip in the destination bar. It streams a long file from storage instead of loading it into the tape/sample buffer.
- The top T1 to T4 buttons choose the active working track. Disc's Output menu separately chooses where the shared deck is heard: Follow Selected moves with the active track, while Pinned keeps it on a fixed track.
- OUT T1 to OUT T4 shows the effective route and its track color. Tap that badge to select the routed track and edit the Grain, Tone, Color, Air, sends, and mixer channel processing Disc.
- Link opens file selection/loading for the shared Disc source. The displayed filename confirms the current file. Eject removes only RUN4's link to that file — it never deletes the original file from Files, iCloud, or an external drive. Position can be scrubbed in the position bar.
- Level sets Disc output. Start chooses the normalized file start point. Loop repeats the file/region when enabled.
- The record itself is a control, not an indicator: drag it to scrub. One full turn is roughly one revolution at 33 1/3 rpm, so a slow hand pitches the material down and a record-speed turn lands near 1x. On a wide layout it fills its column, which is also what makes fine scrubbing possible -- more record under the finger is more resolution per millimetre.
- Disc has its own PLAY and STOP plus the same ALL behavior as the other transport surfaces.

IPHONE



IPAD



Disc on both idioms. A global workspace for long streamed files, outside the four tracks.

PART 04

Devices

The fixed audio chain every track runs through, plus its Mod control sources: Grain, Tone, Color, Air, and Mod.

Grain device

- Grain is an insert device that turns the source into grains, clouds, scans, and delay-like textures. Reach it from a T1–T4 chip, then the sliders icon (Devices), then the Grain tab. Mix defaults to 0, so the device is clean until you blend it in.
- Grain Freeze is the compact final row inside the Grain block. Turn it on to switch to Scan Cloud and stop new audio from replacing the captured grain buffer; the current material keeps playing and can still be scanned, pitched, sprayed, and modulated. Turn it off to resume writing into the buffer.
- Grain: Rate controls grain spawn density; Rate Sync changes it from Free timing to Straight, Dotted, or Triplet divisions. Size controls duration, Contour moves from sharp to soft windows, and Warp bends the read position. Pitch transposes grains, Scale snaps it to Free, Chromatic, Major, Minor, Dorian, Pentatonic, or Fifths, and Detune spreads pitch across grains. Mix blends the original source and grains.
- Motion: Spray randomizes position and stereo spread. Pattern moves from regular pulses toward cloud behavior. Scan travels through recent buffer history and Stretch slows that travel. Scan Mode chooses Stretch, Position, Delay Sync, or Delay Free; Spray Mode chooses stereo Warp or Warp Mono.
- Random: Rate jitters spawn timing, Size jitters duration, Amp randomizes grain level, and Reverse sets reverse-playback probability. Rand Pitch chooses Detune, Random, or Random Mono.
- Engine: Engine selects Live Cloud or Scan Cloud. Feedback chooses input SOS or Grain Output, and SOS controls how much previous material remains in the buffer. Run While Idle keeps the Scan engine moving without active source audio. Save Buffer stores its current contents as reusable material. Delay Time and Delay Beats set free or tempo-synced scan-delay timing.

- Factory presets include cloud and rhythmic starting points plus four source-focused choices: Beat Reinforce, Vocal Halo, Chord Drift, and Bass Shadow. Selecting a preset resets Grain before applying it; user presets can store your own complete setup.

Tone device

- Tone is the filter/resonator device. Reach it from a T1–T4 chip, then the sliders icon (Devices), then the Tone tab. It defaults transparent: lowpass is fully open and Mix is 0 until you engage it.
- Type chooses Lowpass, Highpass, Bandpass, Formant, or Resonator.
- For Lowpass/Highpass/Bandpass/Formant, Cutoff sets the corner or vowel position, Resonance emphasizes the filter, and Mix blends the filter into the track.
- For Resonator, Voices chooses 1 to 16 resonant bands. Root shifts the tuned bank and Scale quantizes it to musical notes. Structure moves from harmonic to stretched or metallic spacing. Detune spreads the individual bands. Resonance sets their emphasis, Decay extends their ringing, and Brightness tilts the bank. Noise excites the resonator internally so it can continue singing over quiet input. Mix blends the resonator.

Color device

- Color adds drive, reduction, and modulation character. Reach it from a T1–T4 chip, then the sliders icon (Devices), then the Color tab. It defaults transparent with Mix at 0.
- Type chooses Saturate, Fold, Bitcrush, or Ringmod.
- Saturate is warm tanh-style saturation. Fold creates richer polynomial wavefolding harmonics. Both use Drive, Tone, and Mix.
- Bitcrush uses Bits and Rate. Lower Bits reduces bit depth; lower Rate reduces sample rate. Tone darkens or opens the result. Mix blends it in.
- Ringmod uses Drive, Frequency, Tone, and Mix. Frequency sets the carrier from low tremolo/ring tones up into metallic sidebands.
- Dynamics works across every Color type. Crossover chooses the low/high split, Focus drives one side harder while protecting the other, Tilt reshapes the wet spectrum, Compress adds level-matched compression, and Gate suppresses low-level material using the clean input as its key.

- Noise adds a character layer after the selected Color process. Amount sets its level, Color moves from darker to brighter noise, and Decay controls how strongly it follows and falls away with the source envelope.

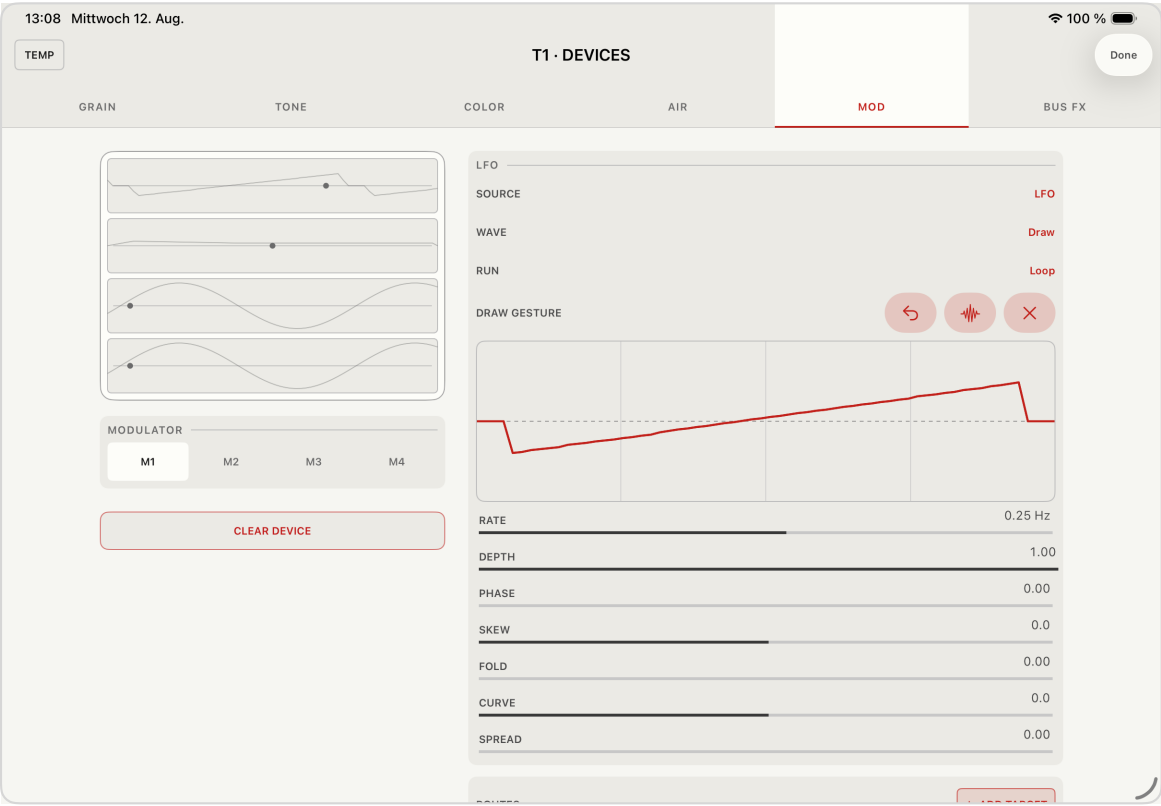
Air device

- Air is the space device: reverb, delay, freeze, repeat color, and broad tonal shaping. Reach it from a T1–T4 chip, then the sliders icon (Devices), then the Air tab.
- Freeze holds the reverb tail. While frozen, the tail sustains as a performance layer.
- Reverb Type chooses Room, Hall, Plate, or Space. Air Mix controls reverb level. Air Size controls tail size/feedback. Damping darkens the reverb.
- Delay Type chooses Digital, Ping-Pong, Tape, or Diffusion. Delay Mix controls delay level. Delay Time sets repeat timing. Delay Feedback controls repeat regeneration.
- Delay Pitch shifts repeats up or down. Chromatic snaps delay pitch motion to semitones; Glide leaves it continuous. Repeat Tone darkens repeats. Saturation soft-clips the delay feedback path.
- Tone is a broad bipolar space tone: negative leans darker/low-pass, positive leans brighter/high-pass.

Modulation and Draw

- Mod is a control device rather than another stage in the audio chain, reached from a T1–T4 chip, then the sliders icon (Devices), then the Mod tab. Each track owns four independent slots, M1 to M4. A slot can be an LFO, Envelope, Random source, Follow source, or Self-Listening source.
- Add Target inside the Mod device to route a source, or long-press any routable parameter and assign it directly. One modulator can drive several targets. Route depth is bipolar, so a source can move one parameter upward and another downward. Multiple sources assigned to the same target are added around its normal base value.
- LFO provides Sine, Triangle, Square, Saw, Sample Hold, Draw, and Steps. Envelope responds to Transport or Note triggers. Random produces stepped or smoothed variation. Follow tracks the selected track's current level value. Self listens to the overall mix energy.

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- To create a touch-drawn modulator, select an M slot, choose LFO as Source, then choose Draw as Wave. Drag from left to right across Draw Gesture to define one bipolar cycle: the center line is zero, above it is positive, and below it is negative. Fast strokes are interpolated so the curve remains continuous.
 - Undo restores the previous drawing, Smooth softens sharp changes, and Clear returns the canvas to zero. The curve can run as a repeating Loop or as One Shot. One Shot can launch from Transport or Note, and Touch Trigger fires it directly from the panel.
 - Steps is the quantised sibling of Draw. Choose LFO as Source and Steps as Wave to get a table of held values instead of a continuous gesture. Steps sets its length from 2 to 32, and the grid shows eight cells per row. Drag across the cells to write the table: the centre line is zero, above it is positive, and below it is negative. One drag can sweep a whole ramp, and the table is stored with the Set like a Draw curve.
 - Any length is allowed, not only powers of two. A table of 13 or 15 steps against a 16-step pattern takes several bars to come back around, which is the cheapest source of movement a sequencer has. Rate, Depth, Phase, Skew, Fold, Curve, and Spread apply to Steps exactly as they do to the other shapes.
 - Rate controls playback speed. Depth scales the result. Phase moves its starting point, Skew changes timing across the cycle, Fold reflects the shape into additional turns, Curve changes its response, and Spread offsets related motion across tracks and slots.
 - Draw curves, source settings, and routes are stored with the current Set.



A real Draw cycle: the painted curve becomes the LFO movement.

PART 05

Sound library

Getting material in, and keeping the states worth returning to.

Sample Browser

- The waveform button in the top row opens Samples. It contains Memories, Recordings, Recently Imported files, and Linked Folders.
- Import copies or links selected audio files through the system picker. Link Folder adds a browsable folder tree using security-scoped access. AudioShare opens AudioShare and returns imported audio to RUN4 when available.
- Tap or the play icon auditions a sample. Stop ends audition. Load Tape loads the selected item into Tape. Load Poly loads it into Poly. Load Disc links it for Disc streaming.
- Recordings can be deleted. Imported history entries can be removed. Samples can be renamed. Linked folders can be unlinked. Missing or unsupported files are marked in the browser.

Memories

- Memories are short captured moments of the overall mix. They are stored as reusable audio snippets and appear in the Samples browser, opened from the waveform button in the top row.
- Memory Listen is a single global capture switch in the Memories section. It captures the post-master mix without loading into or changing a Tape. Tape Listen and Memory Listen share this one internal capture tap: turning Memory Listen on turns any armed Tape Listen off.
- Remember stores the most recent capture window. Auto stores a Memory every 8 seconds while Memory Listen is armed. If the capture buffer has not filled with audio yet, RUN4 reports that instead of creating an empty Memory.
- A Memory can be auditioned, recalled to Tape, or recalled as a variation: Reversed, Pitch +7, Octave down, or Half tempo.

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- The bank is bounded: older unreferenced Memories roll off when full, while a Memory used by any Command Sequence step is kept. Command Memory steps can trigger stored Memories from the tape sequencer and pitch them chromatically.

PART 06

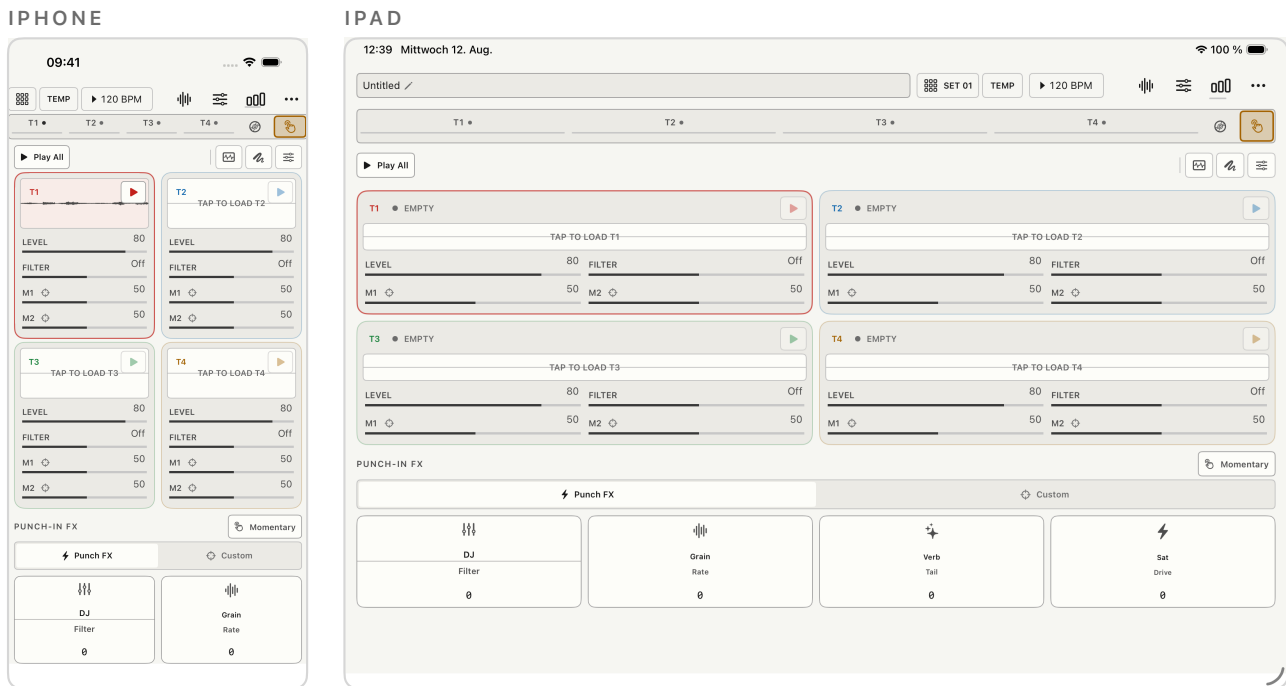
Performance and visualizer

The live stage: Sets, Rune Pad, macros, and the stage graphics.

Performance mode, Sets, Rune Pad, and macros

- Performance opens the dedicated live workspace on iPhone, iPad, and Mac — reach it from the Perform chip at the end of the destination bar. Closing it returns directly to the Tape or Poly view you came from.
- Its four fixed track lanes show T1–T4 waveforms with per-track Play/Stop, Level, DJ Filter, M1, and M2. Keeping every Tape in a stable position makes live switching predictable; phone portrait uses a scrollable stage so the controls retain useful touch length.
- Play All sits at the left of Perform's own action row, beside the visualizer, Rune Pad and settings buttons; it starts or stops every track. Set recall and TEMP are in the app header above, which stays present in Perform. A Set stores the four tracks, material/device state, mixer state, macros, links, and performance setup. Open the Set menu to recall a slot, store the current state, or rename the current Set.
- TEMP is a safe performance layer for temporary parameter changes. Leaving TEMP restores the stored Set; it does not remove an AUv3 instrument or effect.
- Punch-in FX switches between RUN4's immediate global effects and assignable Custom slots. Momentary responds only while a pad is held; Hold latches the effect until it is released.
- The Rune Pad button reveals the XY performance surface and its grouped mapping controls. It appears inline on iPhone and portrait layouts, and as an overlay in wide landscape so the four track lanes remain visible.
- Performance Settings — the sliders icon in Perform's own action row — separates Macros and Track Links into dedicated pages. Macros M1 and onward can target parameters from grouped track/device options; each macro has a Value control from 0 to 1 and an assign menu.

- Macro CC learn maps incoming MIDI CC to a macro. For direct mapping, long-press a routable parameter: Modulation and Macro choices stay first, while MIDI Learn sits at the bottom. RUN4 reuses that parameter's assigned macro or a free macro slot, then learns the next incoming CC. Context menu Clear MIDI CC removes the learned assignment.
- Track Links create reactions between tracks. A source track can point at a target track, choose a link character such as ducking, and set intensity.

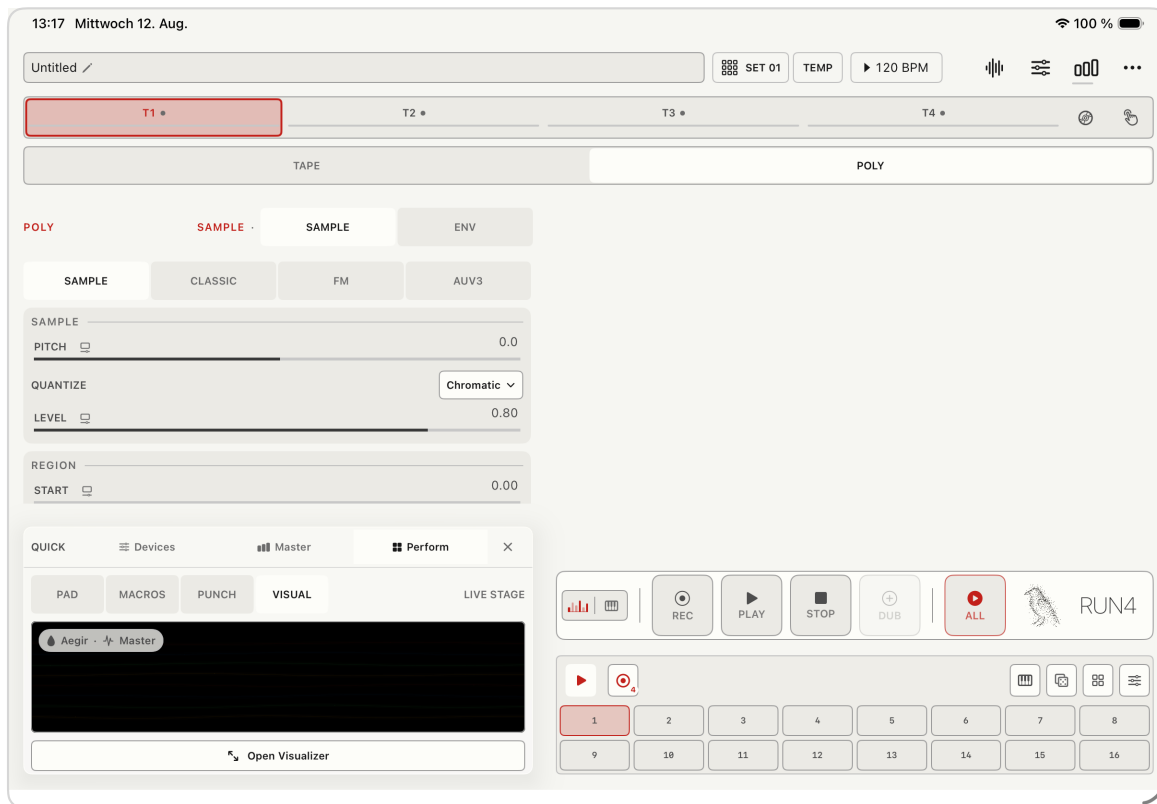


Performance on both idioms. Four track panels with level, filter and macros, over the Punch-in FX row.

Visualizer

- The visualizer turns the four tapes and the master signal into stage graphics. It is read-only: it never changes transport, routing, or any parameter.
- It opens in two places. In Performance, the waveform button in the top toolbar fills the display. In iPad landscape, the Quick dock has a Visual tab holding a live preview; Open Visualizer runs the full stage inside the dock, so tapes, mixer, and transport stay visible and usable around it.
- The visualizer also runs inside the AUv3 plug-in, in the same two steps: Open Visualizer gives you the docked stage in the host's editor, and Full Screen is the step beyond it.

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- From the docked stage, Full Screen promotes the same scene to the whole display. The docked stage stays underneath, so closing full screen returns to the dock you came from, with scene, palette, and source unchanged.
 - Tap once to show or hide the controls. They hide again after about two seconds. While they are visible you can Close, go Full Screen, choose the audio source, choose the scene, switch palette, and open Touch FX.
 - Source selects what drives the graphics: the master signal, or a single tape. Choosing one tape parks the others so the composition stays legible, and that tape's own spectrum drives the scene.
 - Palette switches between Tapes, which uses the four track colours, and Mono, which uses a single hue and lets brightness carry the information.
 - Swipe left or right to change scene. With Touch FX enabled, a one-finger drag belongs to the effects, so scene changes move to a two-finger swipe; the scene menu in the controls always works either way.
 - Touch FX makes the stage a second performance surface. It reuses the Rune Pad engine, so a finger on the stage drives the same effects the Rune Pad does. It is off by default and configured from the Touch FX button.
 - Scenes react to the frequency content of the material, not only its level, so bass, body, and treble drive different parts of a picture. Reduce Motion simplifies every scene: fewer elements, no drift, and no flicker.



The Quick dock on iPad landscape, with the Visual tab holding a live preview. Open Visualizer runs the full stage inside the dock, so tapes, mixer and transport stay usable around it.

Visualizer scenes

- Draupnir. Tape filaments radiating from a common centre, opening and bending with each track.
- Elivagar. Particle streams along curved trajectories sharing one vanishing point.
- Njord. The oscilloscope reimaged: a thick luminous waveform band trailing its own afterimages. Calm material looks calm, loud material erupts.
- Jormungandr. Arcs and points orbiting a centre, tangling as the material builds.
- Freyr. A particle field around a gravity well. Bass swells the haze at the centre and each bar breathes the field outward.
- Aegir. A slow ocean surface.
- Bifrost Gate. A tunnel of radial cells receding into depth, with black wedges opening onto a liquid colour field and a sparse star layer between them.

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- Yggdrasil. A tree whose trunk takes weight from the low bands and whose canopy fills out with the highs.
 - Nidhogg. One closed body of luminous paths, reforming between a dual vortex, a spherical cage, a folded membrane, and a radial burst. A strong attack tears it sideways in raster slices.
 - Jotunheim. Mushroom colonies that emerge, mature, and decay; bass material grows the field.
 - Nornir Loom. An elastic grid of horizontal and vertical lanes that shears, travels, and pulls through a lens at its centre.
 - Idavoll. Four clusters, one per tape.
 - Helheim. Long threads hung across the stage, braided and pinched by coherent crosswinds.
 - Urd. An hourglass of falling particles.
 - Sol. A sun: concentric rings holding the optical centre, with rays that reach out to the edges of the screen.
 - Ran. A folded wave mesh.
 - Mimir. A graph of nodes joined by paths, with shattered plates at node scale.
 - Vetrarbraut. A star road running to a distant point.
 - Fenrir. Nested shells of broken plates that grow, stretch on their own clocks, and dissolve back toward the centre.
 - Nornir. Points joined into shifting constellations.
 - Mjolnir. One ring of particles per frequency band, travelling outward as its band gains energy.
 - Huginn. The RUN4 bird.
 - Galdr. Rune script drawn and redrawn.
 - Midgard. A spectrogram landscape: frequency across, time receding to a horizon, level as height.
 - Thor. A cloud bank with one tape per layer, and lightning that strikes more often as the material gets louder. The bolt takes the colour of the loudest tape.
 - Bifrost. The rainbow bridge, built from the four tape colours blended into a ramp. Each stripe carries one frequency band.

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- Valhalla. A sheet of birds in three depth layers, each bird riding its own part of the spectrum.
 - Skidbladnir. A longship on the swell. The sail's four panels are the four tapes, the shields along the hull follow each track's level, and the hull takes its pitch from the wave beneath it.
 - Langedr. The long-fire of a hall. Four logs, one per tape, glow with their own track; the mids are the flame body and the highs throw sparks. The embers keep breathing when the stage is silent.
 - Niflheim. Frost creeping in from the edges of the glass. The highs grow the crystals, the low end thaws them back, and each seed belongs to one tape.
 - Ginnungagap. The gap where the realms meet: ice reaching in from one side on the highs, fire from the other on the lows, and a churning seam that moves toward whichever side is winning.
 - Nordrljos. Northern lights over a ridge. One curtain per tape, each hung from its own baseline and rippled along its length by the spectrum.

PART 07

Mixing and routing

Levels, sends, the master chain, hardware routing, and getting audio out.

Mixer, sends, and master

- The Master panel shows master level, master dynamics, per-track channels, and AUv3 effect slots — reach it from the bar-chart icon in the top row. On a wide landscape canvas the Mixer uses the full width: the send buses live in BUS FX rather than in a second column beside it.
- Each track has level, pan, and DJ filter controls, plus four hairline A–D send nudges under the filter. Drag one to feed that bus, double-tap it to close the send again. Meters show track and master signal.
- Send A through Send D route every track into four independent send buses. BUS FX inside a track's Devices window is where a whole bus lives: all four track sends, the return level, the shared native effect, and the AUv3 chain in one signal-flow block. It is a live view of the shared bus, not a duplicate plugin chain — Return and FX affect every track feeding it.
- In the AUv3 there is no Devices window, so the plug-in's Mix / Master keeps a full-width Send Bus block under the mixer with the selected bus's return and native effect; its four sends come from the channel strips.
- Long-press a BUS FX Send or Return to assign modulation, a Macro, or Rune Pad control. AUv3 bus editors can open Docked directly below the routing/FX cards, Compact, or Full Screen.
- Mastering uses two rows of five. Signal flow: TONE → DRIVE → GLUE → OTT → WIDTH → LIMITER. LOW/MID/HIGH shape tone, W LOW/MID/HIGH set stereo width, and CEILING sets the limiter wall.
- Choose Neutral, Gentle Glue, Open Highs, Ambient, or Warm Tape from the preset menu. Analyze suggests a starting point without changing the sound; the red rings on GLUE, OTT, and LIMIT show their actual gain reduction.
- AUv3 effects can be loaded directly on the selected bus or on the Master chain, edited when they provide an editor, bypassed, or removed.

Audio interface routing

- Open Settings > Audio > Input & Output Routing to configure a connected class-compliant USB audio interface. RUN4 stores the assignments, but validates them again whenever the hardware route changes.
- Link Audio Inputs in the same routing screen lists audio channels published by compatible Link peers. Enable Link and Audio Sharing, then assign any discovered channel independently to T1–T4. Record and Record & Play capture the selected network stream using the same Free or fixed-Bar length controls as hardware input. The Tape Monitor control listens to the selected Link source live; otherwise it monitors that Tape's hardware input.
- Hardware Route leaves that Tape on its physical input assignment. If a saved Link peer disappears, RUN4 preserves its name for reconnection and safely uses the hardware route until the channel returns.
- Tape Inputs assigns an independent mono channel or stereo pair to T1–T4. This allows simultaneous captures such as input 1 to T1, input 2 to T2, input 3 to T3, and input 4 to T4. Detect Input Channels briefly opens the record-capable route when iOS does not expose the full input count while idle.
- Hardware Outputs sends each Tape either to Master, Off, a physical mono output, or a physical stereo pair. Master Output can stay on the normal system stereo pair, move to another physical pair, become mono, or be disabled.
- A Tape assigned directly to hardware is removed from the dry Master path so it is not heard twice. Its send-bus contribution still returns through Master, keeping shared effects on the main mix.
- Examples: route T1–T4 to mono outputs 1–4 for a four-channel mixer, or to stereo pairs 1–2, 3–4, 5–6, and 7–8 for four independent stereo stems.
- If an assignment is invalid, the interface disconnects, or the multi-output renderer cannot start safely, RUN4 restores the normal stereo Master automatically. Physical channel counts and multi-output availability depend on the connected interface and iOS.

Export and bounce

- Open Export & Bounce from the top ... menu. It is a separate full workspace so delivery controls do not lengthen or clutter the live mixer.

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- Export records the live output in real time. Select Master and/or T1 to T4 Stem, then choose Record or Record & Play. Stop Export finishes and opens sharing for the created files.
 - Offline Bounce renders without playing in real time. Select Master and/or supported track stems, choose Duration, then Bounce Offline.
 - Offline stem support depends on the source and state. The app shows a detail line for unsupported options instead of silently failing.
 - Use live export for performance captures and gestures. Use offline bounce for fixed-length renders, clean stems, and transfer to another DAW.

PART 08

Sync and control

Playing RUN4 from, and alongside, everything else.

MIDI settings

- Reach these from the ellipsis menu, then Settings, then the MIDI page. Input chooses All MIDI sources or a specific source. Refresh Devices rescans available sources. Bluetooth MIDI opens the iOS Bluetooth MIDI connection sheet.
- Poly Channels assigns a separate MIDI channel to each of T1 to T4 for note input. Control Channel chooses which channel receives global controls, or All.
- Program Change enables MIDI program changes for Set recall. Macro CC enables MIDI CC control of learned macros.
- MIDI Mappings holds a Transport section above the macro banks: press Learn next to Play All, Stop All, Record, or Dub and then send the button or pad you want to use. Learn accepts a note or a CC, so pad controllers work as well as dedicated transport buttons, and a pad mapped here drives transport instead of playing a note. Record and Dub follow the selected track and are ignored on a Disc deck. Unlike macros, these mappings belong to the controller rather than the project, so they stay put when you switch projects. The section's toggle disables controller transport without deleting what you learned.
- Save Current as Layout stores the controller's transport buttons and Macro CC positions for reuse across projects. Macro targets stay with each project, so one controller layout can play different project-specific macro assignments.
- MIDI Clock In/Out, Ableton Link, Sync Quantize, Rec Grid, and the Metronome all live in Timing & Sync -- open it here or long-press the BPM display anywhere in RUN4. MIDI Clock In lets RUN4 follow external MIDI clock and transport; MIDI Clock Out sends RUN4 clock to other devices; Sync Quantize queues synced actions (Record, Overdub, Play All) to the next beat while Link or MIDI Clock is running.
- Rec Grid quantizes Record, Overdub, and Play All to RUN4's own tempo grid when neither Link nor MIDI Clock is running -- pick a musical value (8 bars down to 1/16) and a punched action waits for the next grid line before it starts; Free launches immediately, as before. The first take defines the grid, so later records on other tracks lock to it.

- The waveform-plus control inside Tape's transport box opens the capture panel for Record or Record & Play, Free or a fixed 1/2/4/8-bar length, Count-in Off or 1 Bar, the Metronome, and its continuous level fader. A Count-in shows READY 4 · 3 · 2 · 1 before capture. Fixed captures then show bars and remaining time in the waveform HUD and close automatically; Record & Play immediately loops the new take.
- The Metronome clicks that same grid: quarter notes at the project tempo, accenting beat 1 of each bar, so a quantized record has an audible reference. Timing & Sync provides its continuous level control. In the standalone app its isolated output is not captured into Tape or Export. Inside the AUv3 the click leaves on RUN4's output bus, so the host can record it; it is still added after RUN4's own internal Tape capture point.
- Long-press the Metronome to choose when it clicks after capture begins: Always, Rec (only while recording or overdubbing), Play (only during playback), or Rec+Play (during either recording or playback). The separate Count-in always supplies its own four preparation clicks without changing this choice.
- Virtual Input exposes RUN4 as a MIDI input endpoint for other apps on the device.

Ableton Link and AUv3

- Project BPM is available from the top mode row. Drag the BPM chip to change tempo, tap it to switch the same compact chip to fine +/- 0.1 steps, or long-press it to open Timing & Sync.
- Ableton Link syncs tempo and transport with other Link-enabled apps on the local network. Timing & Sync shows Link status and transport state.
- RUN4 updates Link tempo from the project BPM and can react to Link start/stop when enabled in the Link settings sheet.
- When Link Audio is enabled, RUN4 can share the master Mix plus four post-fader track streams with compatible Link Audio peers that request them on the same local network.
- RUN4 can also receive compatible Link Audio channels directly into any Tape. Choose the source under Settings > Audio > Input & Output Routing; the normal Tape recording controls then capture that stream without an AUv3 routing utility. Tap Monitor on that Tape to hear the assigned Link channel live without recording it twice.
- RUN4 can host AUv3 instruments in Poly and AUv3 effects in the mixer/master path.

- Hosted AUv3 instruments and effects receive RUN4's tempo, musical position, time signature, and transport start/stop. Tempo-driven plugins such as gates, filters, and grooveboxes can therefore follow the same RUN4 transport used for Tape and fixed-bar recording.
- AUv3 editor choices are Docked, Compact, and Full Screen where supported. Docked keeps the plugin in the current RUN4 workspace; rotation preserves the selected device tab instead of returning to the first page.
- RUN4 can also run inside compatible hosts as AUv3 FX, Source, and Instrument components. The plug-in carries the same workspace as the app, including the full-width Mix / Master page with Clarity, live meters, per-channel sends and the send-bus block, a Master page in the Quick panel, and the visualizer.
- The host's own preset menu holds RUN4's Sets. Recalling one is a real parameter move: it voices all four device chains AND the tape controls -- speed, pitch, keylock, reverse, gain, loop fades, wow and flutter, wear and tape type -- so the Tape page shows what you recalled. Recorded audio, the loop region and the transport are never touched, and the Init Set puts every one of those controls back to neutral.
- A few things stay standalone-only because a plug-in cannot do them: hosting other AUv3 instruments or effects, file export and bounce, and the AudioShare import (its SDK needs app-level APIs that iOS does not give an extension). Use Import in the plug-in instead -- AudioShare appears there as a file provider.

Mac and AUv3 editor note

- RUN4 is the full native app on iPad, iPhone, and Apple-silicon Mac. The standalone Mac window uses the complete RUN4 interface and can be resized freely; the layout follows the real window size.
- RUN4 AUv3 uses that same complete interface — Settings, playback modes, Scenes, Tape, Bloom, Poly and Command — without a reduced plug-in view or a second zoom overlay.
- An AU host owns the embedded editor window and its resize handles. RUN4 asks it for the largest available editor viewport, then adapts the real interface to every size the host provides.

PART 09

Reference

Working habits, key commands, and what to do when something misbehaves.

Performance tips

- For fast resampling, enable Listen, perform across tracks, then Capture into an empty Tape. This avoids microphone bleed and keeps the capture internal.
- For evolving pads, use Tape with Bloom, Grain Mix, Tone Resonator, and Air Freeze. Keep Color subtle unless you want obvious grit.
- For rhythmic glitch, use Command with Jump, Stutter, Gate, Retrigger, and Nudge. Use Probability below 1 for variation.
- For playable instruments, load a short sample into Poly, set Start/Length/Loop/XFade, then shape Attack/Decay/Sustain and play by MIDI.
- For long backing layers, use Disc instead of Tape so the file streams from storage and does not need to live in the tape buffer.
- For mix clarity, keep device Mix controls low at first, then raise only the effect that contributes to the part.

Keyboard shortcuts

- Connect a physical keyboard to use Poly as a computer MIDI keyboard. In Poly, press M or tap Keys to turn QWERTY MIDI on or off. Keys On is shown while performance keys are active.
- Command- and Control-modified keys keep their normal system and text-editing behavior. Leave a text field before performing so its typing focus does not take the keyboard.

WITH QWERTY MIDI ON

A W S E D F T G Y H U J	One chromatic octave from Poly's Base note
K	
Z / X	Octave down / octave up

Shift + key	Stronger note velocity
M	Turn QWERTY MIDI off again
WITH QWERTY MIDI OFF	
1 to 4	Select track
[/]	Previous / next track
T / P / D / F	Tape / Poly / Disc / Perform
R	Start or stop the selected Poly pattern while Poly is active
Space	Start or stop global playback

Troubleshooting

- No sound: check selected track, transport state, Material mode, tape/sample/disc source, track level, master level, and whether device Mix controls are unexpectedly at 0 or 1.
- Recorded over a tape by mistake: open the Undo menu in the top bar and choose Undo Record. The previous take's audio also remains in the Sample Browser under recordings, so it can be reloaded manually even later.
- Capture is silent: enable Listen first, wait for the listen buffer to contain audio, avoid capturing while conflicting record/overdub actions are running, and confirm the source is audible in RUN4.
- MIDI does not appear: use Settings > MIDI > Refresh Devices, choose Input All, check the source app/device channel, or connect through Bluetooth MIDI.
- Hardware keyboard does not play Poly: select Poly, press M or tap Keys to enable QWERTY MIDI, and leave any active text field. With Keys On, A through K are notes rather than navigation shortcuts.
- AUv3 instrument does not load or reports that it could not connect safely: RUN4 automatically retries one recoverable graph connection. If it still fails, refresh the AUv3 list, confirm the plugin supports a Music Device component, then remove and reload it.
- A Docked or Compact AUv3 editor is blank: close and reopen the editor or try Full Screen. Some plugins advertise only specific view configurations even though their audio engine is running.

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- A tempo-based AUv3 does not follow transport: confirm RUN4 transport is running and the plugin's own sync/host-transport option is enabled.
 - Audio dropouts: reduce heavy AUv3 plugins, lower dense Grain/Bloom/Command activity, stop unused tracks, freeze/bounce complex material, or export stems.
 - A linked file is missing: open Samples, check Linked Folders, re-link the folder or import the file again.
 - Offline bounce has disabled stems: read the detail line next to the track. Some live or plugin-driven states may require live export instead.